

C optimize.c X

C: > Users > Afroz > Desktop > EXP9 > C optimize.c

```
1  #include <stdio.h>
2  #include <string.h>
3
4  struct op
5  {
6      char l;
7      char r[20];
8  } op[10], pr[10];
9
10 int main()
11 {
12     int i, j, k, n, z = 0;
13     char temp;
14     char *p;
15
16     printf("Enter the number of values: ");
17     scanf("%d", &n);
18
19     for (i = 0; i < n; i++)
20     {
21         printf("left: ");
22         scanf(" %c", &op[i].l);
23
24         printf("right: ");
25         scanf(" %s", op[i].r);
26     }
27
28     printf("\nIntermediate Code\n");
29
30     for (i = 0; i < n; i++)
31         printf("%c = %s\n", op[i].l, op[i].r);
32
33     /* Dead Code Elimination */
34     for (i = 0; i < n - 1; i++)
35     {
36         temp = op[i].l;
37
38         for (j = 0; j < n; j++)
39         {
40             p = strchr(op[j].r, temp);
41
42             if (p)
43             {
44                 pr[z].l = op[i].l;
45                 strcpy(pr[z].r, op[i].r);
46                 z++;
47                 break;
48             }
49         }
50     }
51 }
```

```

51
52     pr[z].l = op[n - 1].l;
53     strcpy(pr[z].r, op[n - 1].r);
54     z++;
55
56     printf("\nAfter Dead Code Elimination\n");
57
58     for (k = 0; k < z; k++)
59         printf("%c = %s\n", pr[k].l, pr[k].r);
60
61     /* Common Expression Elimination */
62     printf("\nEliminate Common Expression\n");
63
64     for (i = 0; i < z; i++)
65     {
66         for (j = i + 1; j < z; j++)
67         {
68             if (strcmp(pr[i].r, pr[j].r) == 0)
69                 pr[j].l = '\0';
70         }
71     }
72
73     printf("\nOptimized Code\n");
74
75     for (i = 0; i < z; i++)
76     {
77         if (pr[i].l != '\0')
78             printf("%c = %s\n", pr[i].l, pr[i].r);
79     }
80
81     return 0;
82 }
83

```

```

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP9
$ ./optimize
Enter the number of values: 4
left: a
right: b+c
left: d
right: a*2
left: e
right: b+c
left: f
right: e+1

Intermediate Code
a = b+c
d = a*2
e = b+c
f = e+1

After Dead Code Elimination
a = b+c
e = b+c
f = e+1

Eliminate Common Expression

Optimized Code
a = b+c
f = e+1

```